

# Reactionless Compliance: Which Redundancy Resolution Should a Teleoperated Free-Floating Servicer Use?

[TODO: five names]

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**Abstract**—A robot arm on a free-floating spacecraft pushes its own base around whenever it moves. A 7-joint arm has spare freedom that can be spent on cancelling that push, on keeping the arm’s shape predictable for the operator, or on neither. We compare five redundancy resolution schemes in a momentum-conserving MuJoCo simulation, on three servicer sizes (30, 300 and 3000 kg base, 12.5 kg arm) and two task types (position only, and position plus orientation). With a position-only task the reactionless schemes cut peak base attitude drift by a factor of six on the 300 kg servicer ( $0.20^\circ$  against  $1.22^\circ$ ) with tracking still under 0.2 mm. With a full 6-D task the freedoms run out, and drift can only be bought with tracking error: on the 30 kg servicer, staying inside a  $5^\circ$  pointing limit costs about 80 mm of hand error. On the 3000 kg servicer the choice of scheme changes base drift by less than  $0.01^\circ$ , so the spare freedom is better spent on the operator. A contact-speed map from ISO/TS 15066 gives 0.28 to 0.76 m/s depending on servicer size and posture. A participant study of operator predictability is specified but not yet run.

## I. INTRODUCTION

On-orbit servicing puts a human operator in control of an arm whose base is not bolted down. Momentum is conserved, so every hand motion rotates the spacecraft, which moves antennas and sensors off their targets [1], [2]. Reaction null-space (RNS) control uses arm redundancy to move the hand without rotating the base [3], [4]. Separately, human–robot interaction work shows that operators do better when the arm’s configuration is repeatable, which null-space impedance toward a rest posture provides [5], [6]. Both draw on the same small pool of spare joints.

**Gap.** Reactionless control is studied for what it does to the spacecraft, and null-space compliance for what it does to the person. We found no study that puts both on the same arm and asks what the operator gives up for a quiet base, or how the answer changes with spacecraft size.

**Contributions.** (i) A degree-of-freedom argument, checked numerically, for when a quiet base is free and when it is not. (ii) A five-scheme, three-size, two-task benchmark with a verified momentum-conserving model. (iii) The trade-off curve between hand error and base drift. (iv) A contact-speed map per servicer size. (v) A ready-to-run protocol for the operator study.

## II. RELATED WORK

[TODO: P2: condense P2\_Literature\_Survey.xlsx to about half a page.] The generalised Jacobian [2] gives hand motion

of a free-floating arm from joint rates alone. RNS [3] and its flight test on ETS-VII [4] show zero-reaction manoeuvres. Damped least squares [7] and gradient projection [8] are the standard redundancy baselines. Effective mass at the contact point [9] links arm posture to collision severity, the basis of the power-and-force limiting mode in ISO/TS 15066 [10], [11].

## III. MODEL

The system is a rigid base on a free joint with a 7-revolute arm (z-y-z-y-z-y-z, reach 1.09 m), no gravity, simulated in MuJoCo [12] at 0.5 ms with RK4. With base twist  $v_b$  and joint rates  $\dot{q}$  the mass matrix splits as

$$M = \begin{bmatrix} H_b & H_{bm} \\ H_{bm}^\top & H_m \end{bmatrix}, \quad H_b v_b + H_{bm} \dot{q} = 0 \quad (1)$$

when starting from rest. So  $v_b = V \dot{q}$  with  $V = -H_b^{-1} H_{bm}$ , and the hand twist is

$$\dot{x} = (J_m + J_b V) \dot{q} = J^* \dot{q}. \quad (2)$$

Let  $W$  be the three angular rows of  $V$ . The base does not rotate exactly when  $W \dot{q} = 0$ . (Using the null space of the angular block of  $H_{bm}$  instead is only correct when the base frame is at the system centre of mass.)

**Verification.** Under random joint commands for 10 s the base-row momentum stays below  $10^{-6}$  in SI units on all three servicers, and  $J^* \dot{q}$  and  $W \dot{q}$  match the simulated hand and base velocities to  $10^{-6}$ .

**Freedom count.**  $W$  has rank 3, leaving a 4-dimensional reactionless subspace. A position-only task needs 3, so  $3 + 3 = 6 \leq 7$ : exact, with one joint to spare. A 6-D task needs  $6 + 3 = 9 > 7$ : no exact solution exists. Both statements are checked by rank tests on all sizes.

## IV. SCHEMES

All schemes output joint rates, tracked by joint velocity servos and limited to 2 rad/s by scaling the whole vector.  $q_0 = -k_p(q - q_{\text{rest}})$ .

- 1) **DLS:**  $\dot{q} = J^{*\top} (J^* J^{*\top} + \lambda^2 I)^{-1} \dot{x}$ .
- 2) **Gradient projection:** pseudo-inverse plus manipulability gradient in the null space.
- 3) **Null-space impedance:** pseudo-inverse plus  $q_0$  in the null space.
- 4) **RNS:** task and  $W \dot{q} = -k_a \theta_b$  together,  $\theta_b$  the base attitude error.

TABLE I  
POSITION-ONLY TASK: HAND ERROR / PEAK BASE DRIFT

| Scheme      | 30 kg           | 300 kg      | 3000 kg      |
|-------------|-----------------|-------------|--------------|
| DLS         | 0.14 mm / 17.9° | 0.12 / 1.22 | 0.12 / 0.040 |
| Grad. proj. | 0.08 / 16.4     | 0.11 / 1.08 | 0.12 / 0.037 |
| Impedance   | 0.08 / 19.7     | 0.11 / 1.23 | 0.12 / 0.040 |
| RNS         | 47.0 / 3.97     | 0.16 / 0.22 | 0.10 / 0.016 |
| Imp.+RNS    | 49.7 / 1.94     | 0.16 / 0.20 | 0.11 / 0.027 |

5) **Impedance + RNS**: as 4, with  $q_0$  in whatever freedom is left.

For the position-only task, 4 and 5 stack  $[J_{pos}^*; W]$  and solve exactly. For the 6-D task they solve

$$\dot{q} = (J^{*\top} J^* + \beta W^\top W + \gamma I)^{-1} (J^{*\top} \dot{x} - \beta k_a W^\top \theta_b + \gamma q_0), \quad (3)$$

where  $\beta$  sets how much hand accuracy is traded for a quiet base.

**Joint limits.** Every scheme is wrapped in the same guard. A joint within 0.05 rad of its limit ( $\pm 2.9$  rad on the roll joints,  $\pm 2.0$  rad on the pitch joints) that is still commanded outward is locked, and the remaining joints re-solve. The hand task is weighted above the reactionless term, so whatever the locked joints can no longer cancel goes into base motion rather than hand error. In the scripted benchmark only pure RNS reaches a limit (36% of the run on the 300 kg servicer); no run exceeds one.

## V. EXPERIMENT

The hand follows a 0.10 m circle in the x-z plane of the inertial frame, twice, 15 s per loop, starting and ending at rest; the 6-D task adds a 0.25 rad tool tilt. A scripted operator (feed-forward plus proportional correction) stands in for the human so that runs are repeatable. Design: 5 schemes  $\times$  2 tasks  $\times$  3 sizes, plus  $\beta \in [10^{-2}, 10^6]$  for schemes 4 and 5 in 6-D. Measures: RMS hand position error, peak base attitude drift, joint return error  $\|q_{\text{end}} - q_{\text{rest}}\|$ , smallest singular value of  $J^*$ . All 84 runs were stable.

## VI. RESULTS

**Position-only task (Table I).** On the 300 kg servicer the reactionless schemes cut peak drift five to sixfold with tracking still under 0.2 mm. With scheme 5 the base returns to its start attitude (0.01°). Plain RNS ends 0.13° off because one joint drifts onto its limit, and the others end 0.4 to 0.7° off. Drift is not exactly zero as the freedom count predicts; we attribute the remainder to velocity-servo lag, which the attitude feedback corrects only after it appears. [TODO: confirm by rerunning with ideal joint rates.] On the 30 kg servicer the reactionless solution holds a joint on its limit for 24 to 44% of the run, so the hand falls 47 to 50 mm behind and the base still turns 2 to 4°: a quiet base is feasible in principle and not on this path.

**6-D task (Fig. 1).** There is no free lunch. On the 300 kg servicer RNS moves from 1.7 mm error and 1.5° drift ( $\beta = 10^{-2}$ ) to 56 mm and 0.32° ( $\beta = 10^2$ ), and at  $\beta = 10^6$  the base is still and the hand barely follows (114 mm). On the

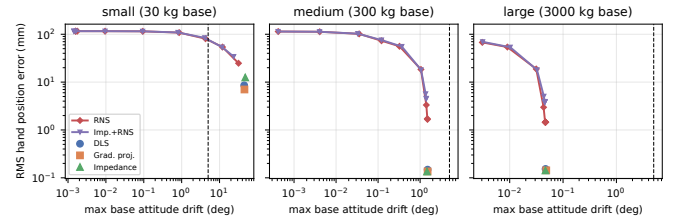


Fig. 1. 6-D task. Each line is one reactionless scheme as  $\beta$  goes from  $10^{-2}$  to  $10^6$ . Dashed: 5° pointing limit.

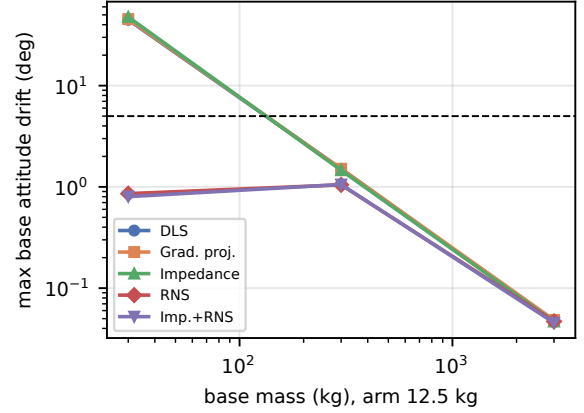


Fig. 2. Peak base drift against base mass, 6-D task,  $\beta = 10$ .

30 kg servicer schemes 1 to 3 drift 45 to 48°, far outside a 5° limit; getting inside it needs  $\beta \geq 1$  and costs 81 to 84 mm.

**Servicer size (Fig. 2).** Drift falls about 30 times for each tenfold step in base mass. At 3000 kg every scheme stays under 0.05° and the reactionless term buys nothing measurable in 6-D, while costing 1.3 to 3.7 mm of tracking.

**Joint drift (Fig. 3).** Null-space impedance returns the arm to its rest shape (0.010 rad position-only, 0.021 rad 6-D); DLS ends 0.05 to 0.20 rad away. Pure RNS is the worst by far (3.1 rad position-only, on a joint limit for half the run), because it spends all spare freedom on the base. Adding impedance to RNS recovers part of that position-only (1.18 rad) and most of it in 6-D (0.14 against 0.97 rad), with slightly less base drift in the position-only task (0.20 against 0.22°).

**What a level base costs in reach (Table II).** Restricted to the joint motions that do not turn the base, the hand Jacobian of the 300 kg servicer at rest has singular values 0.36, 0.10 and 0.08, against 0.22 to 0.74 unrestricted: one direction stays about four times freer than the other two. Over 6000 random postures the smallest value has median 0.01 and never exceeds 0.10, so no starting posture removes the limit, and from 0.08 at rest it falls to 0.007 after 0.15 m forward and below 0.0001 after 0.14 m sideways, a dynamic singularity [1]. Reach is therefore bought with drift. The price is set by base inertia: at rest the worst hand direction turns the base by 9.9°, 0.76° and 0.03° per 10 cm on the 30, 300 and 3000 kg servicers. Holding the hand still leaves the spare joints 10 to 20% of their

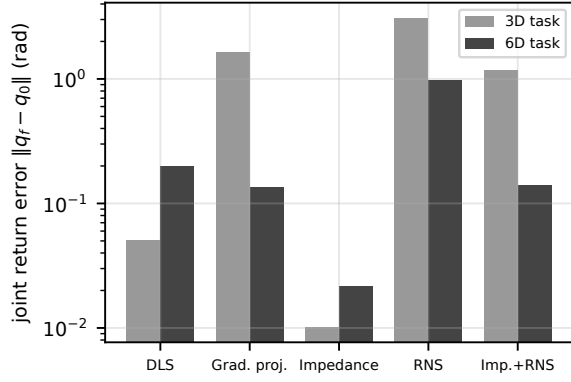


Fig. 3. Joint return error after two loops, 300 kg servicer.

TABLE II  
STRAIGHT-LINE HAND REACH FROM REST AGAINST BASE DRIFT ALLOWED (HAND FIRST,  $w = 2$ )

| Drift allowed | 30 kg | 300 kg           | 3000 kg |
|---------------|-------|------------------|---------|
| 0.5°          | 0.15  | 0.25 (0.11–0.37) | 0.61    |
| 1°            | –     | 0.31 (0.17–0.48) | –       |
| 2°            | 0.18  | 0.46 (0.27–0.70) | 0.61    |
| none          | –     | 0.59 (0.27–1.20) | –       |

Mean over  $\pm x, \pm y, \pm z$  in metres (range); arrival within 5 mm.

authority over the base, so the base only partly re-levels at the goal (1.52° on arrival, 0.79° after 8 s, one joint on its limit). The drift does not fully come back with the hand either. From the earlier rest pose it did (0.00° after 0.3 m out and back). From the ready pose 0.30° is left, with the joints 1.24 rad from where they started, and three laps of a square leave 0.67, 0.11 and 0.14°. Base tilt is only roughly a function of where the hand is.

**Sizing (Fig. 4).** Nine base masses from 30 to 3000 kg give a design chart for this 12.5 kg arm: the full 0.61 m straight-line reach needs about 1700 kg at a 0.5° limit and about 950 kg at 2°; a 100 kg servicer keeps 0.17 to 0.23 m. Holding the base with reaction wheels instead does not get cheaper on a bigger bus: the momentum they must carry is  $h = Kv$  for hand speed  $v$ , with  $K$  between 8 and 11 kg m at rest across all nine masses (median 8, 90th percentile 13 over 2000 postures at 300 kg), because it is the arm’s momentum and not the base’s. At the 0.1 m/s used here that is 0.8 to 1.1 N m s. Base drift scales with the bus; wheel demand scales with the arm and with hand speed. Against catalogue small-satellite wheels (vendor datasheet values: 0.5 N m s at 25 mN m, 1.0 N m s at 60 mN m, 12 N m s at 200 mN m) the momentum is affordable and the torque is not: with  $K = 8$  kg m a 1 N m s wheel holds the base up to  $v = h/K \approx 0.12$  m/s, but its 60 mN m lets the hand accelerate at only  $\tau/K \approx 0.008$  m/s<sup>2</sup>, about 13 s to reach 0.1 m/s, and 4 s even on the 12 N m s wheel. Wheels can hold the base for slow free-space motion; they cannot for a contact transient, which is where the arm’s own null space is needed.

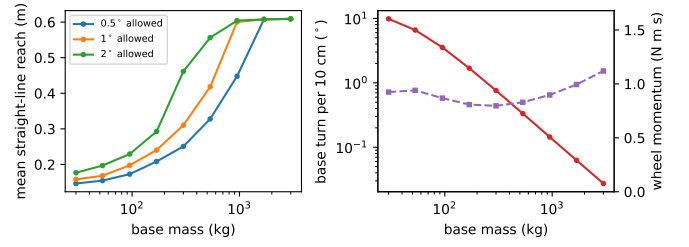


Fig. 4. Left: mean straight-line reach against base mass for three drift limits. Right: base turn per 10 cm of hand travel (solid) and the momentum reaction wheels would carry to cancel it at 0.1 m/s (dashed).

## VII. CONTACT SAFETY

The mass a person feels at the hand along direction  $u$  is  $m_{\text{eff}} = (u^T J M^{-1} J^T u)^{-1}$  [9]. We take the conservative case of locked joints, so the whole servicer is behind the hand. With the transient chest limits of ISO/TS 15066 [10] ( $F_{\text{max}} = 280$  N,  $k = 25$  kN/m, body mass 40 kg [TODO: P4: check against Annex A]),  $v_{\text{max}} = F_{\text{max}}/\sqrt{\mu k}$  with  $\mu$  the reduced mass.

Fig. 5: 0.39 to 0.76 m/s on the 30 kg servicer, 0.30 to 0.44 on 300 kg, and 0.28 to 0.29 on 3000 kg, where the base is so heavy that posture stops mattering. The task’s peak hand speed is 0.063 m/s, well inside all three. A floating base makes contact gentler, and a light one more so.

## VIII. OPERATOR STUDY

[TODO: Not run. No participant data exists.] Protocol: within-subject, each participant flies one loop under all 5 schemes  $\times$  2 tasks by gamepad, order shuffled and scheme hidden. After each trial: a 1–7 predictability rating; after each scheme: NASA-TLX [13]. Objective measures are the same as Section V. Hypothesis: schemes with null-space impedance (3, 5) are rated more predictable than their counterparts (1, 4). [TODO: Fig. 4, statistics, ethics statement.]

## IX. DISCUSSION AND CONCLUSION

The spare joints of a 7-joint arm are worth different things on different spacecraft. For position-only work inside the level-base envelope, impedance plus RNS is the best of both: quiet base, repeatable arm. For 6-D work on a small servicer the operator must be given, or must choose, a point on the trade-off curve. On a large servicer reaction is negligible and the freedom should go to the operator. For teleoperation the same trade-off sets the work envelope: the space an operator can be offered is a choice of how far the base may tilt, and because that tilt is roughly repeatable it can be shown to the operator before the move rather than discovered during it. Away from the edge, whether a goal is accepted did not depend on where the hand started: 36 goals at 0.6, 0.9 and 1.15 of the edge got the same answer from rest and after a 0.15 m detour (72 of 72). Close to the edge it did: on 60 goals between the probed lines, across four detours, the answer changed on 1 of 80 at 0.95 of the edge, 4 of 80 at 1.00 and 18 of 80 at 1.05. A larger detour reaches further in: after 0.3 m detours in five directions the answer changed on 1 of 100 goals at 0.6 of the edge, 5

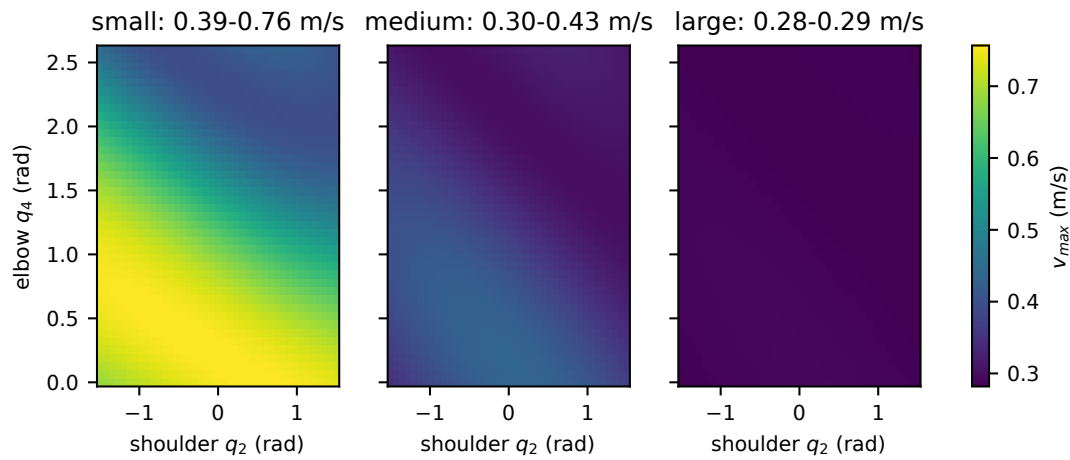


Fig. 5. Permitted hand speed over shoulder and elbow angle.

at 0.8, 6 at 0.9, 24 at 1.00 and 36 at 1.05. So an envelope probed from rest along 98 directions is a guide and not the gate. From rest it agreed with a rehearsal of the move on 271 of 300 goals placed near its edge; from elsewhere the rehearsal has to decide.

**Limitations.** One arm, one path, one rest posture; gains tuned by hand on one servicer; scripted operator only; no flexible appendages, joint friction or sensor noise; initial momentum zero; safety values are for one body region and a locked-joint worst case.

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